

5 (**short**) Figure 2 shows a device for controlling the water level in a small reservoir. The gate is rectangular, 0.5 m wide (into the page), and the other dimensions are as shown in the figure. The density of water may be assumed to be 1000 kg m^{-3} .

(a) Calculate the mass M such that the gate begins to open when the water level is 2 m.

[5]

(b) It is suggested that, as an alternative method of controlling the water level at 2 m, a hole could be made in the gate, and a valve fitted to control the flow. The smallest possible size valve is to be used and the maximum expected water flow rate into the reservoir is $0.001 \text{ m}^3 \text{ s}^{-1}$. State, with your reasons, where the valve should be placed, and calculate its effective flow area.

[5]

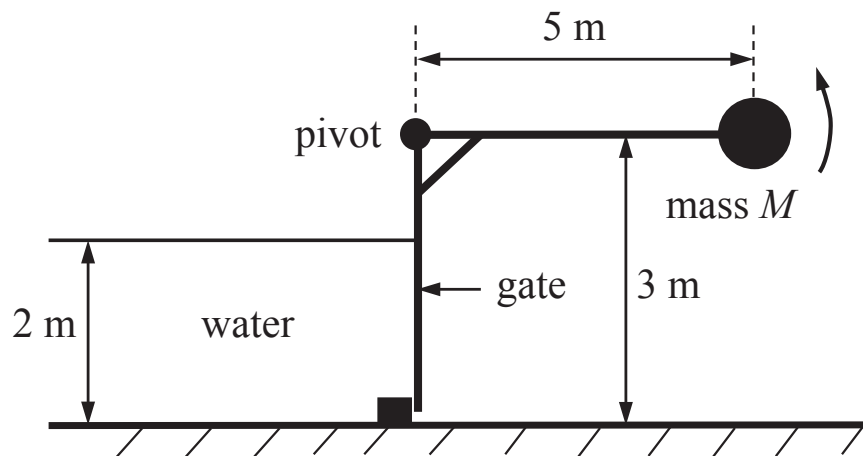


Fig. 2